Most potent GO Biological Process Gene Set Sorted by BMD 25% Relative Deviation at the 5th Percentile

DTTID: 108-020-004-000-5
Study Number: 108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley SD

Test Compound: 1,4-Dichlorobenzene
CAS Number: 106-46-7
DTXSID: DTXSID1020431

Date: 28 July 2025
Time: 12:19:10 PM

Female

Tissue	Category Name	Input Genes/Platform Genes in Gene Set	% Gene Set Coverage	BMD_{rd25} [a] at the 5th Percentile (ppm)	$BMDL_{rd25}$ at 5th Percentile - $BMDU_{rd25}$ at 5th Percentile (ppm)	Genes with Changed Direction Up	Genes with Changed Direction Down
Liver	**GO:0006805 xenobiotic metabolic process**	**20/59**	**33.9**	**87.6** [b]	**59.6 - 136.7**	**18**	**2**
Kidney	GO:0044770 cell cycle phase transition	4/58	6.9	121.0	42.2 - 652.8	1	3
Heart	GO:0044770 cell cycle phase transition	5/58	8.6	141.4	55.9 - 292.9	0	5
Lung	GO:0007059 chromosome segregation	9/41	22.0	208.6	55.4 - 1082.9	0	9
Ovary	GO:0001667 ameboidal-type cell migration	5/55	9.1	223.1	90.9 - 432.8	4	1

[a] rd25 = 25% relative deviation

[b] tPOD (transcriptional Point of Departure); determined based on the lowest GO BP BMDrd25 at the 5th Percentile.

Active Genes:

GO:0006805: Ugt2b1;Gstp1;Gstm2;Gsta4;Gsta1;Gsta3;Gsta2;Gsta5;Fmo1;Cyp3a2;Cyp3a23-3a1;Cyp2j4;Cyp2e1;Cyp2c6;Cyp2c11;Cyp2b2;Cyp2b1;Cbr1;Acaa1b;Acaa1a
GO:0044770: Hspa8;Cks2;Ccnb2;Ccnb2-ps2
GO:0044770: Usp22;Cks2;Ccnb2;Ccnb2-ps2;Ccna2
GO:0007059: Ttk;Smc2;Nuf2;Mad2l1;Kif18b;Dlgap5;Ctcf;Cenpw;Cenpf
GO:0001667: S100a9;Nrp1;Kdr;Cxcl12;Apoa1

Annotations

Official active gene symbols from the Rat Genome Database are shown in the "Active Genes" section above. Definitions of Gene Ontology terms were provided by the Gene Ontology Resource (http://geneontology.org/) and are included below.
GO process description version: https://doi.org/10.22427/NTP-DATA-002-00600-0002-000-0 V06132025

GO Biological Process Definitions:

GO:0006805 xenobiotic metabolic process: The chemical reactions and pathways involving a xenobiotic compound, a compound foreign to living organisms. Used of chemical compounds, e.g. a xenobiotic chemical, such as a pesticide.
GO:0044770 cell cycle phase transition: The cell cycle process by which a cell commits to entering the next cell cycle phase.
GO:0007059 chromosome segregation: The process in which genetic material, in the form of chromosomes, is organized into specific structures and then physically separated and apportioned to two or more sets. In eukaryotes, chromosome segregation begins with the condensation of chromosomes, includes chromosome separation, and ends when chromosomes have completed movement to the spindle poles.
GO:0001667 ameboidal-type cell migration: Cell migration that is accomplished by extension and retraction of a pseudopodium.